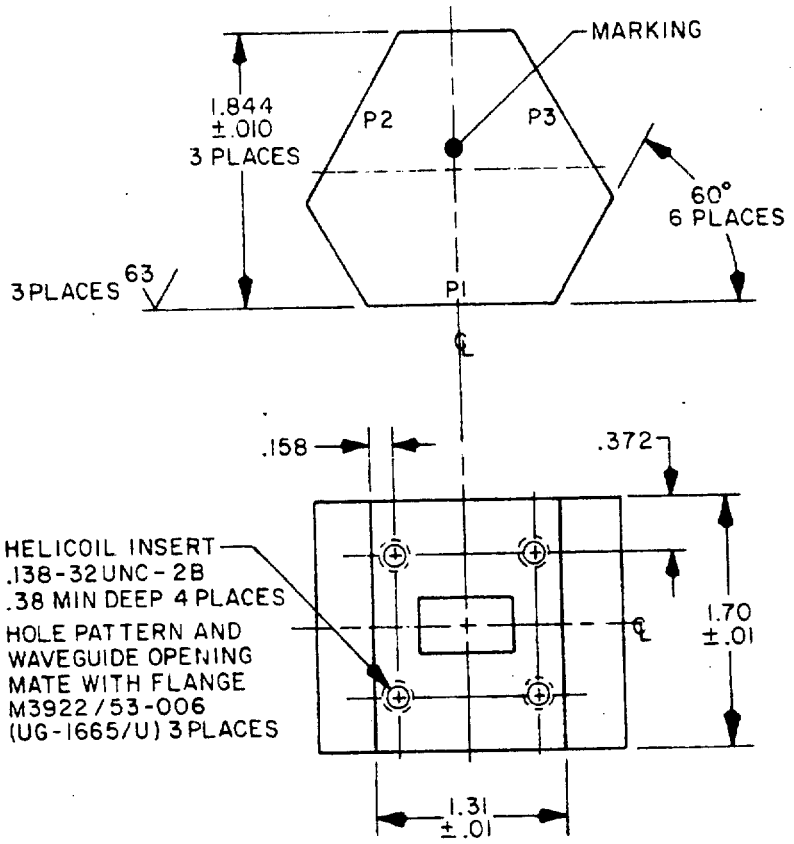
MILITARY SPECIFICATION SHEET

CIRCULATORS, RADIO FREQUENCY, WAVEGUIDE, THREE PORTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the circulators described herein shall consist of this document and the latest issue of specification MIL-C-28790.



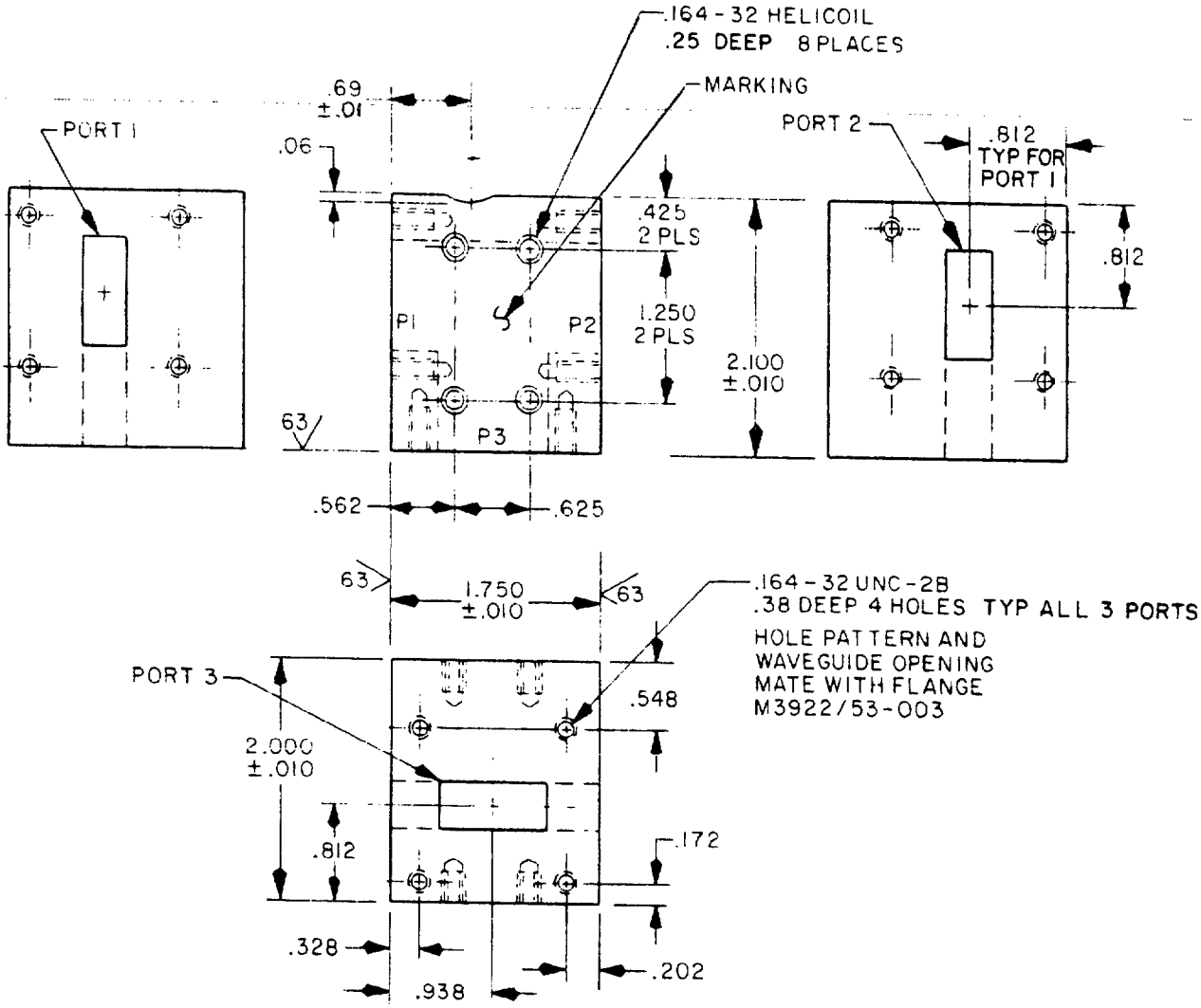
INCHES	MM
.010	.25
.138	3.51
.158	4.01
.372	9.45
.38	9.7
1.31	33.3
1.70	43.2
1.844	46.84

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Port 1 shown; ports 2 and 3 are the same as port 1.

FIGURE 1. Dimensions and configuration for part number M28790/A-001 and -002.

FSC 5985

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INCHES	MM	INCHES	MM	INCHES	MM
.010	.25	.425	10.80	1.250	31.75
.06	1.5	.548	13.92	1.50	38.1
.164	4.17	.562	14.27	1.750	44.45
.172	4.37	.625	15.88	2.000	50.80
.202	5.13	.69	17.5	2.100	53.34
.328	8.33	.812	20.62		
.38	9.7	.938	23.83		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Port 3 shown; ports 1 and 2 are the same as port 3.

FIGURE 2. Dimensions and configuration for part number M28790/6-003.

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ENGINEERING DATA:

Frequency range: See table I.
Temperature range: See table I.
Power rating: See table I.
Circulation: 1 → 2 → 3 → 1.
Weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.
Isolation: See table I.
Insertion loss: See table I.
VSWR: See table I.
Shock: Test condition K.
Temperature cycling: Test condition A, method 107, of MIL-STD-202.
Vibration: Test condition A.
Pressurization: Leakage shall not exceed 0.005 cubic inch per minute when pressurized to 20 psig with air and all ports sealed. This test shall be performed after the vibration test.
Marking: See figures 1 and 2.
Part number: M28790/6-(dash number from table I).

TABLE I. Electrical requirements.

Dash number	Frequency range	Isolation	Insertion loss	VSWR	Temp range		Power rating	Material	Weight (approx)
					Operating	Non-operating			
	GHz	Min	Max	Max			Min		Lbs
-001	10.8-18.0	10 dB 10.8-12.4 GHz 17 dB 12.4-18.0 GHz	1 dB 10.8-12.4 GHz 0.5 dB 12.4-18 GHz	2.0:1 10.8-12.4 GHz 1.3:1 12.4-18 GHz	0° to +75°C	-62° to +75°C	100 W (avg)	AL	0.75
-002	↓	↓	↓	↓	↓	↓	↓	Cu	2.5
-003	9.5-10.6	20 dB	0.3	1.2:1	-55° to +85°C	-55° to +125°C	225 W (avg)	AL	0.6

NOTE: Circulators covered by this specification sheet are substituted for the following manufacturer's part numbers. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	Manufacturer's code	Manufacturer's part number	FSN
-001	05869	723357-1	
-001	14135	Ku130LY	
-001	49956	CKuH105	
-002	05869	723357-2	
-002	14135	Ku201LY	
-003	06401	7910704	
-003	51027	B3X23313	

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force - 11
DLA - ES

User activities:
Army -
Navy - OS, SH, MC
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-28790/6 Circulators, Radio Frequency, Waveguide, Three Ports

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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